

Differentiability and Review on Domain and Interval Notation (Homework)

The graphs of functions are given. Use these graphs to write the domain and range of each function in words, set notation, and interval notation.

① Domain

Words:

Set Notation:

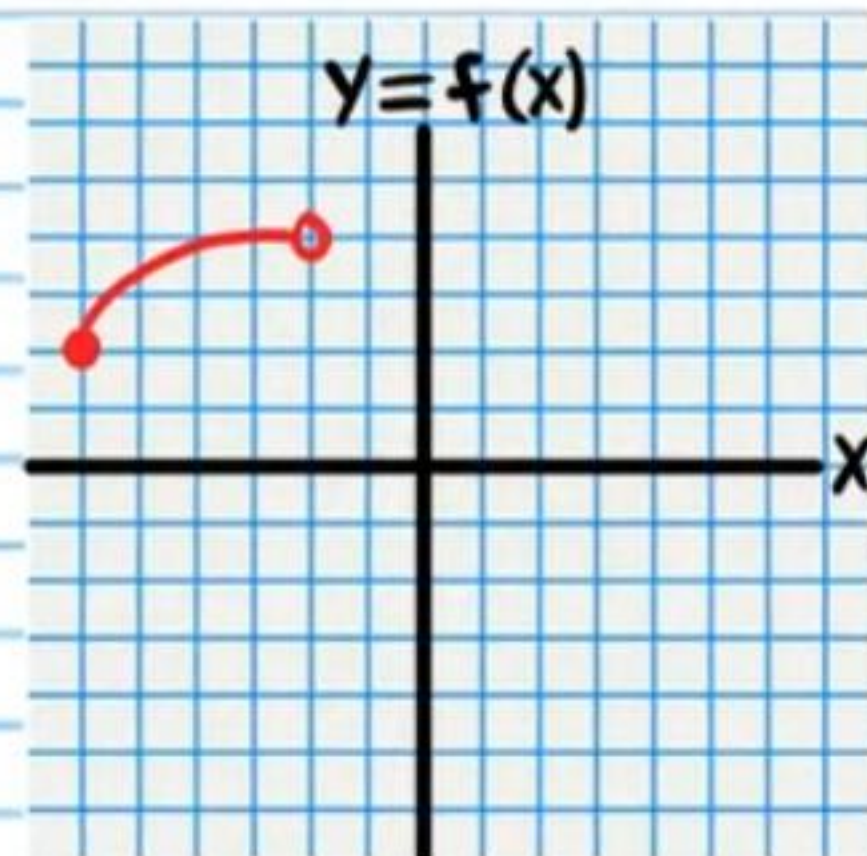
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



② Domain

Words:

Set Notation:

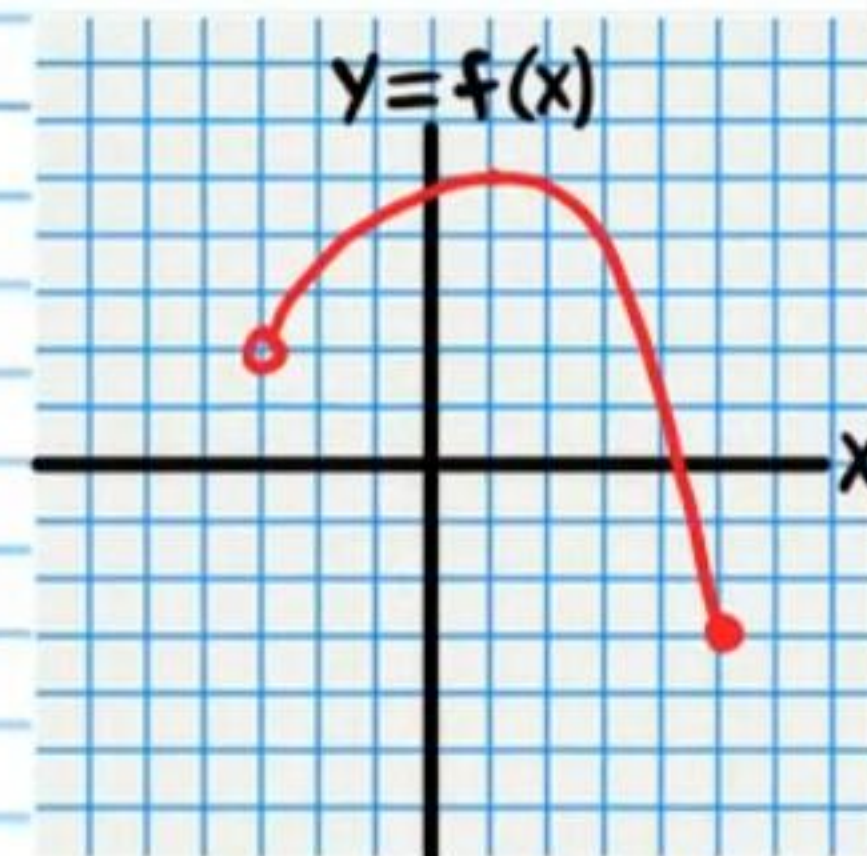
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



③ Domain

Words:

Set Notation:

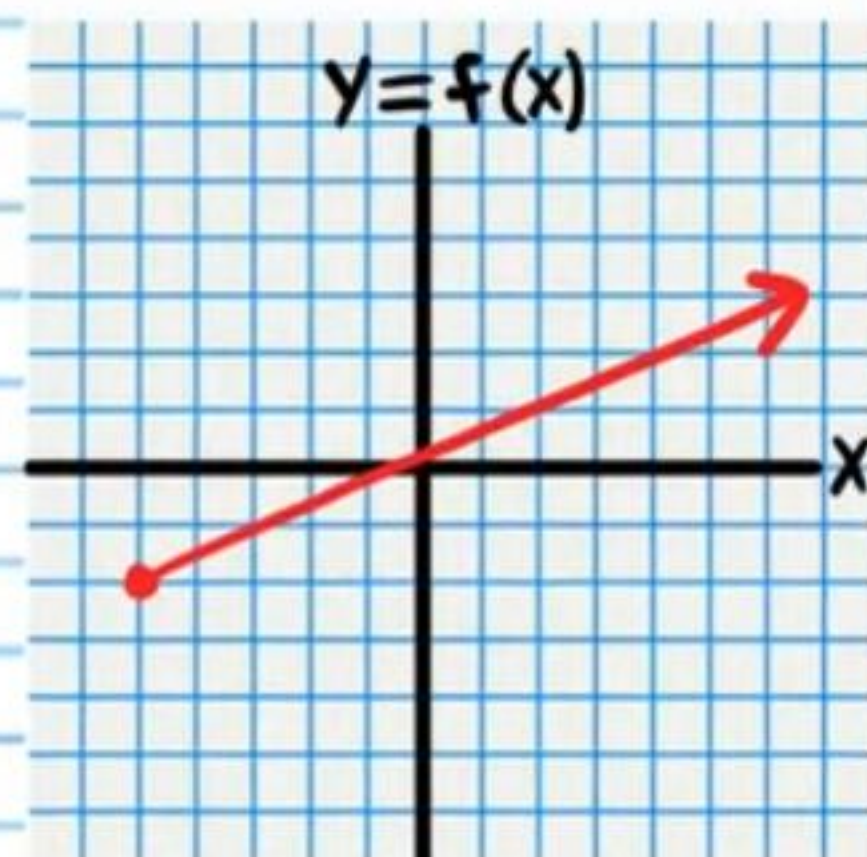
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



④ Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

⑤ Domain

Words:

Set Notation:

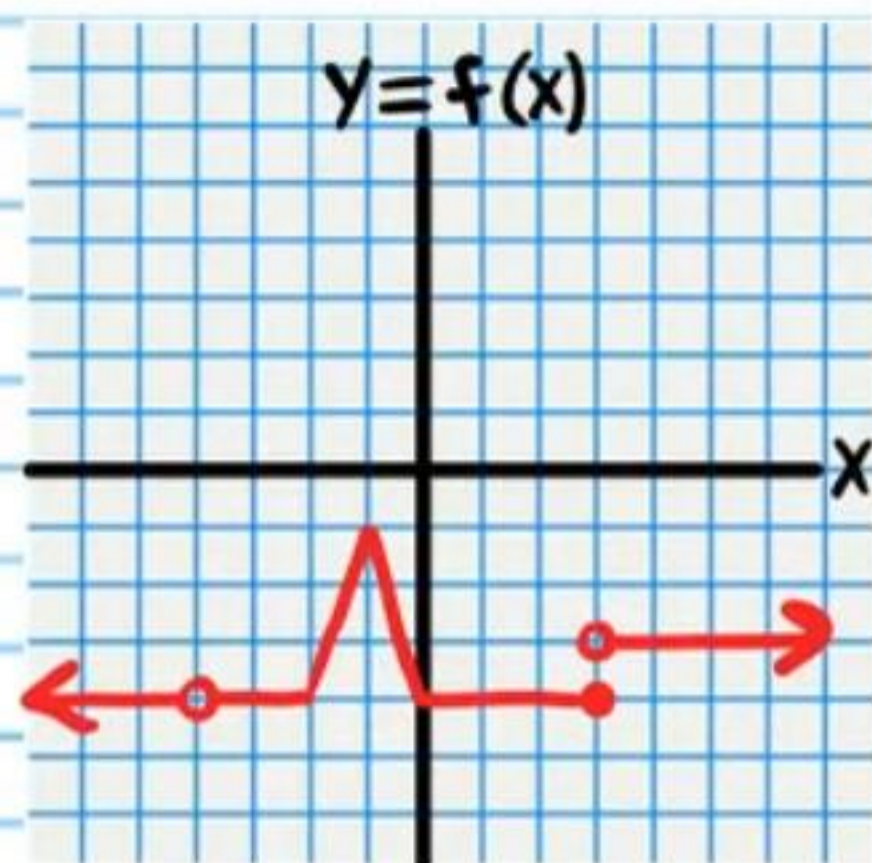
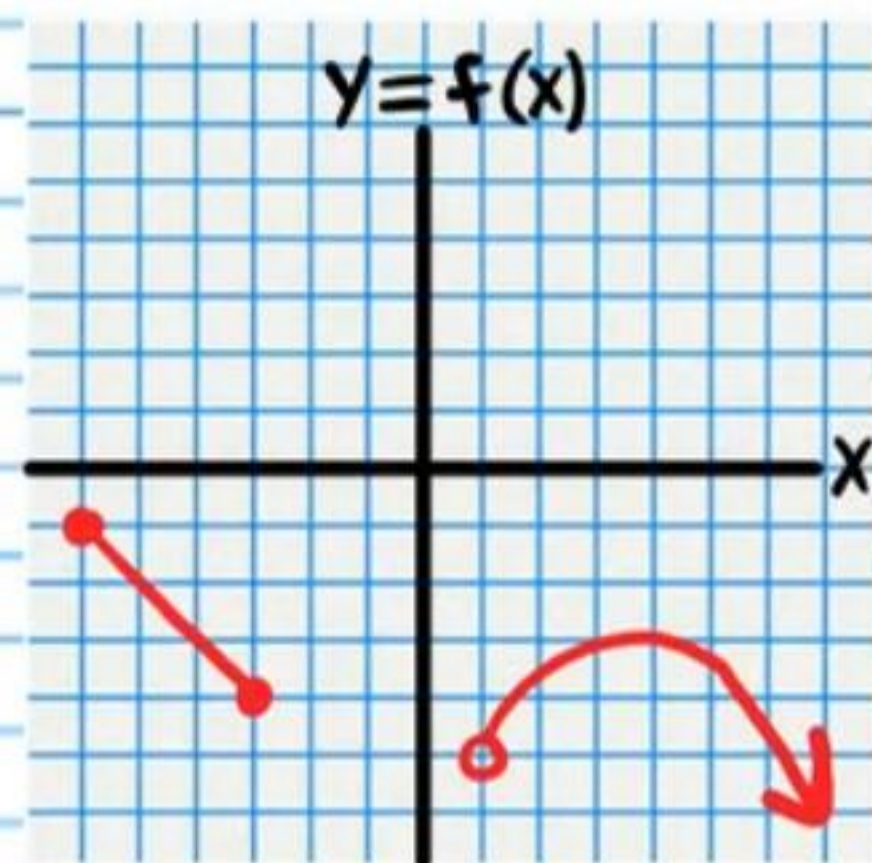
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑥ Domain

Words:

Set Notation:

Interval Notation:

Range

Words:

Set Notation:

Interval Notation:

⑦ Domain

Words:

Set Notation:

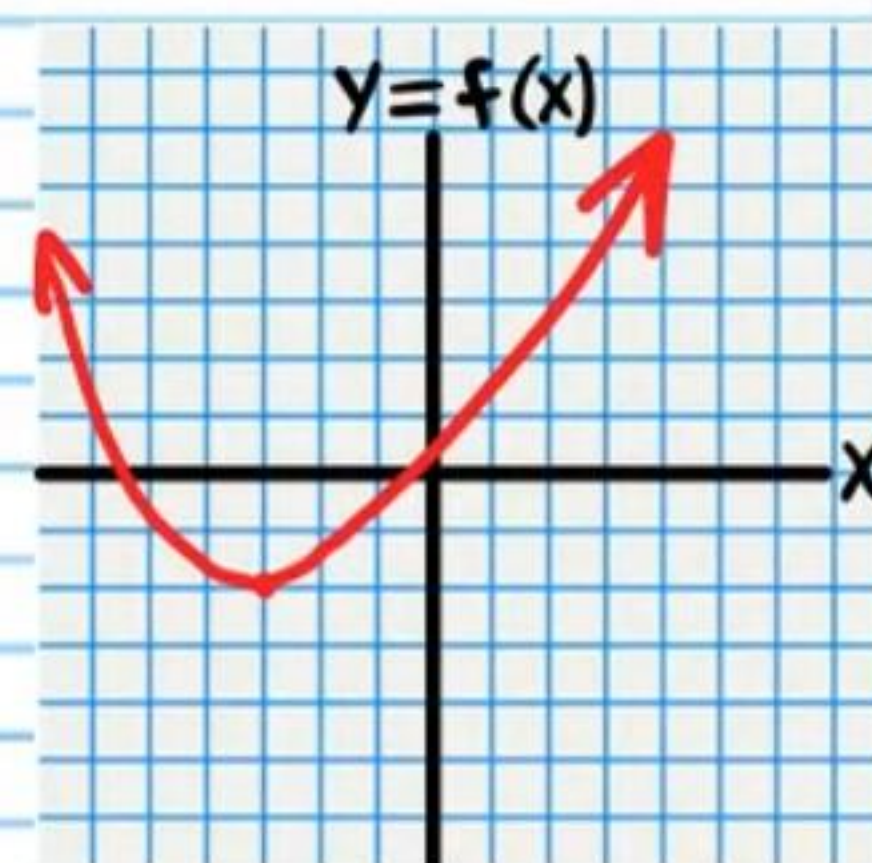
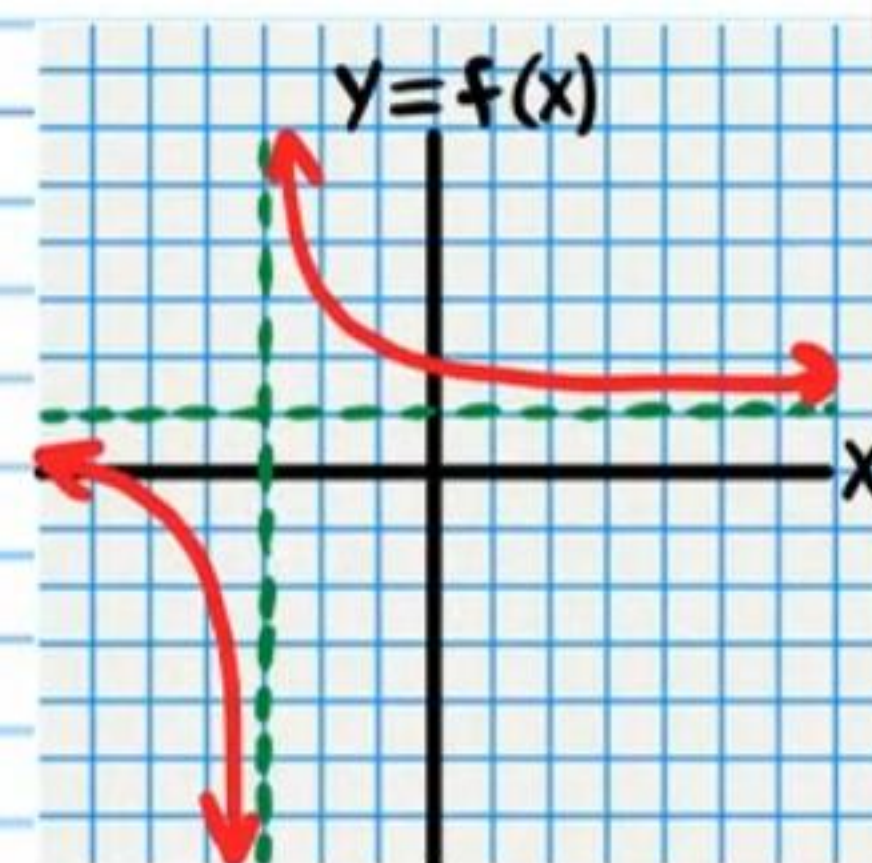
Interval Notation:

Range

Words:

Set Notation:

Interval Notation:



⑧ Domain

Works:

Set Notation:

Interval Notation:

Range

Works:

Set Notation:

Interval Notation:

Find the domain of each function below.

$$\textcircled{9} f(x) = \frac{1}{3x-1}$$

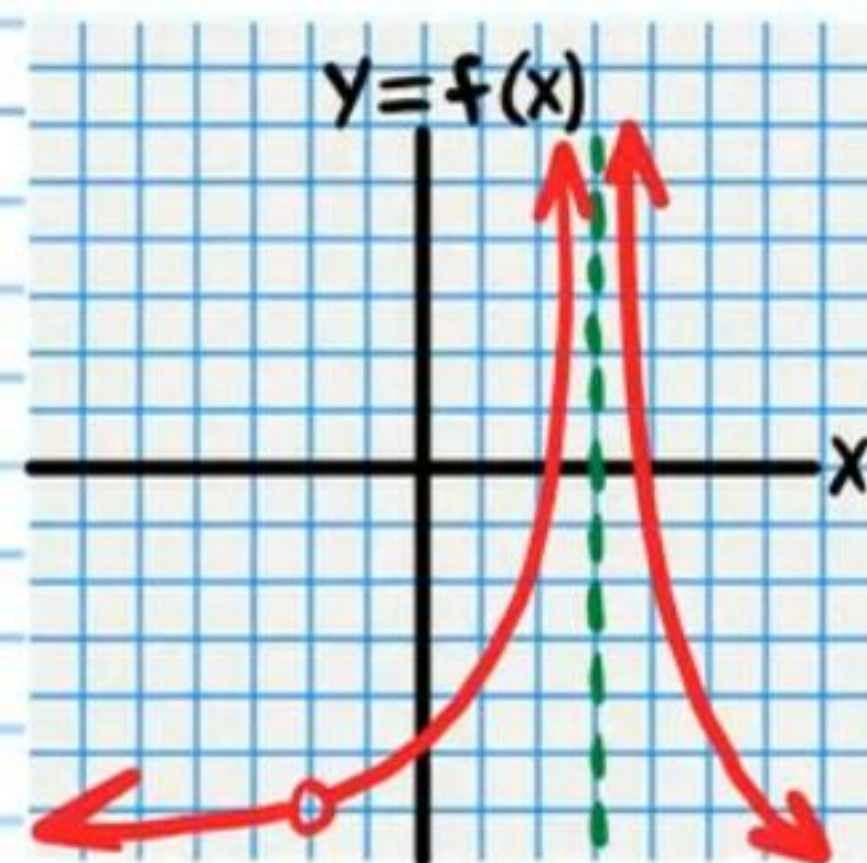
Reasoning: A fraction with zero as its denominator is undefined, so _____ cannot be zero.

Domain

Works:

Set Notation:

Interval Notation:



$$\textcircled{10} g(x) = 2\sqrt{x-6}$$

Reasoning: The square root of a negative number is an imaginary number, so _____ must be greater than or equal to zero.

Domain

Works:

Set Notation:

Interval Notation:

$$\textcircled{11} \quad h(x) = \frac{\log_8(-\frac{x}{5}+3)}{4} + 6$$

Reasoning: Both zero and negative numbers produce an undefined logarithmic expression, so _____ must be greater than zero.

Domain

Words:

Set Notation:

Interval Notation:

$$\textcircled{12} \quad f(x) = -2x^2 + x - 100$$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

Domain

Words:

Set Notation:

Interval Notation:

$$\textcircled{13} \quad f(x) = |2x+22| + \frac{1}{4}$$

Reasoning: The function has no square roots, fractions, logarithms, or other things that might restrict its domain.

Domain

Words:

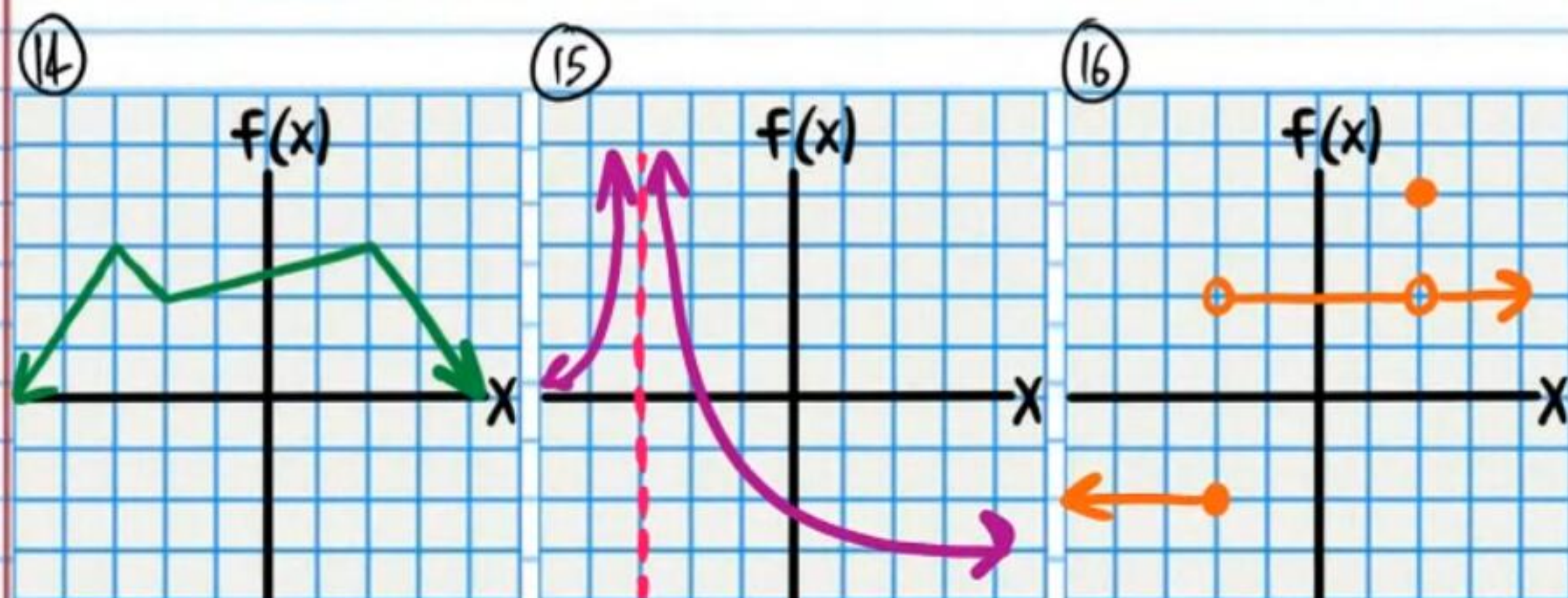
Set Notation:

Interval Notation:

For problem #'s 14-18,

a) Write all x-values where there is a discontinuity.

b) Write all x-values where the function is not differentiable.



a)

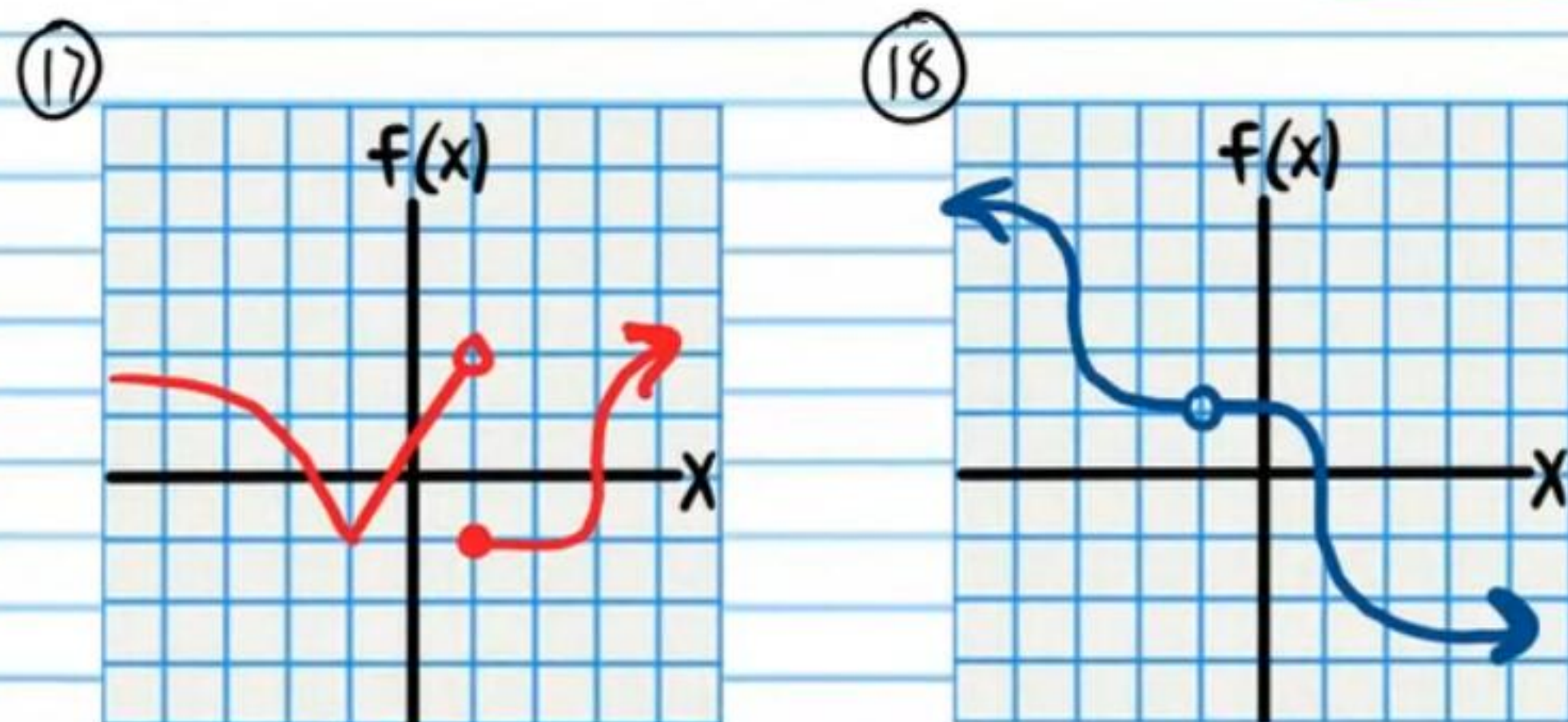
b)

a)

b)

a)

b)



a)

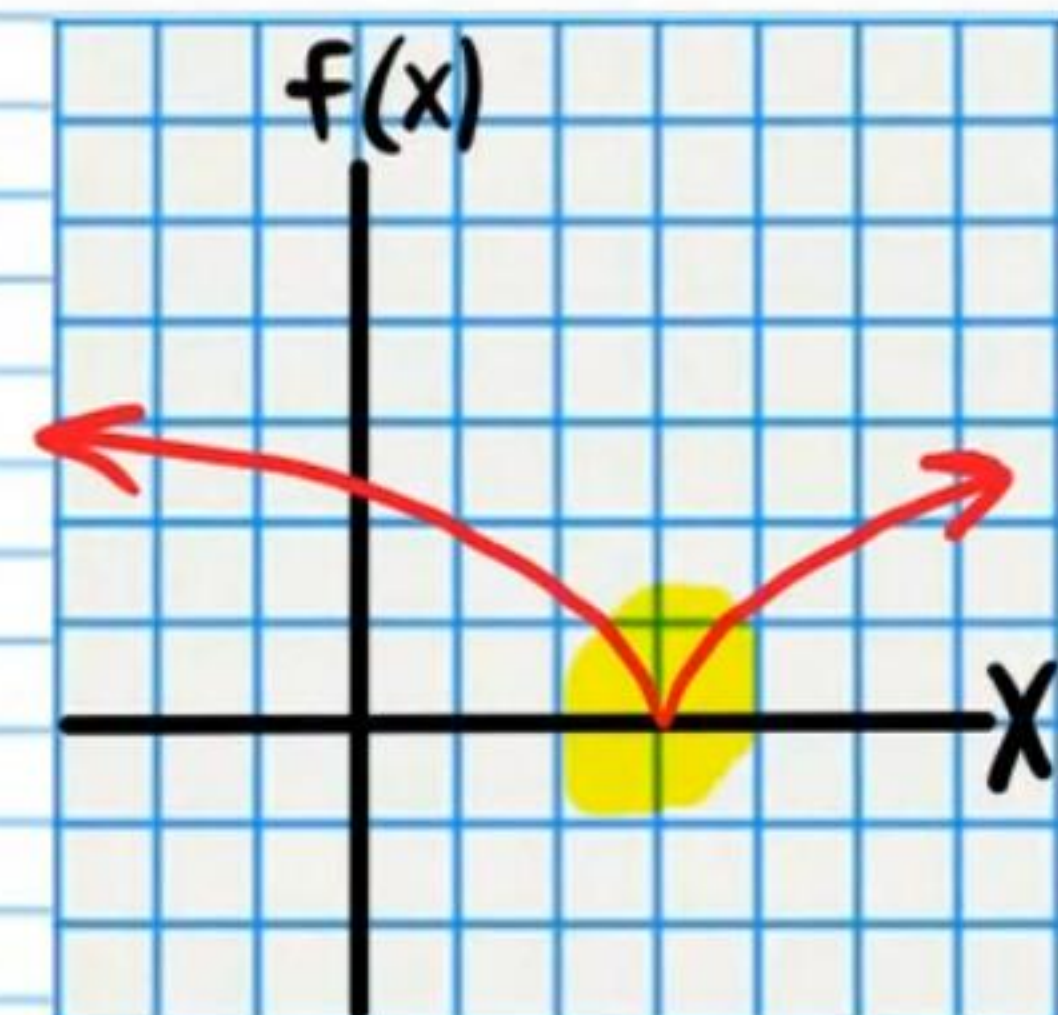
b)

a)

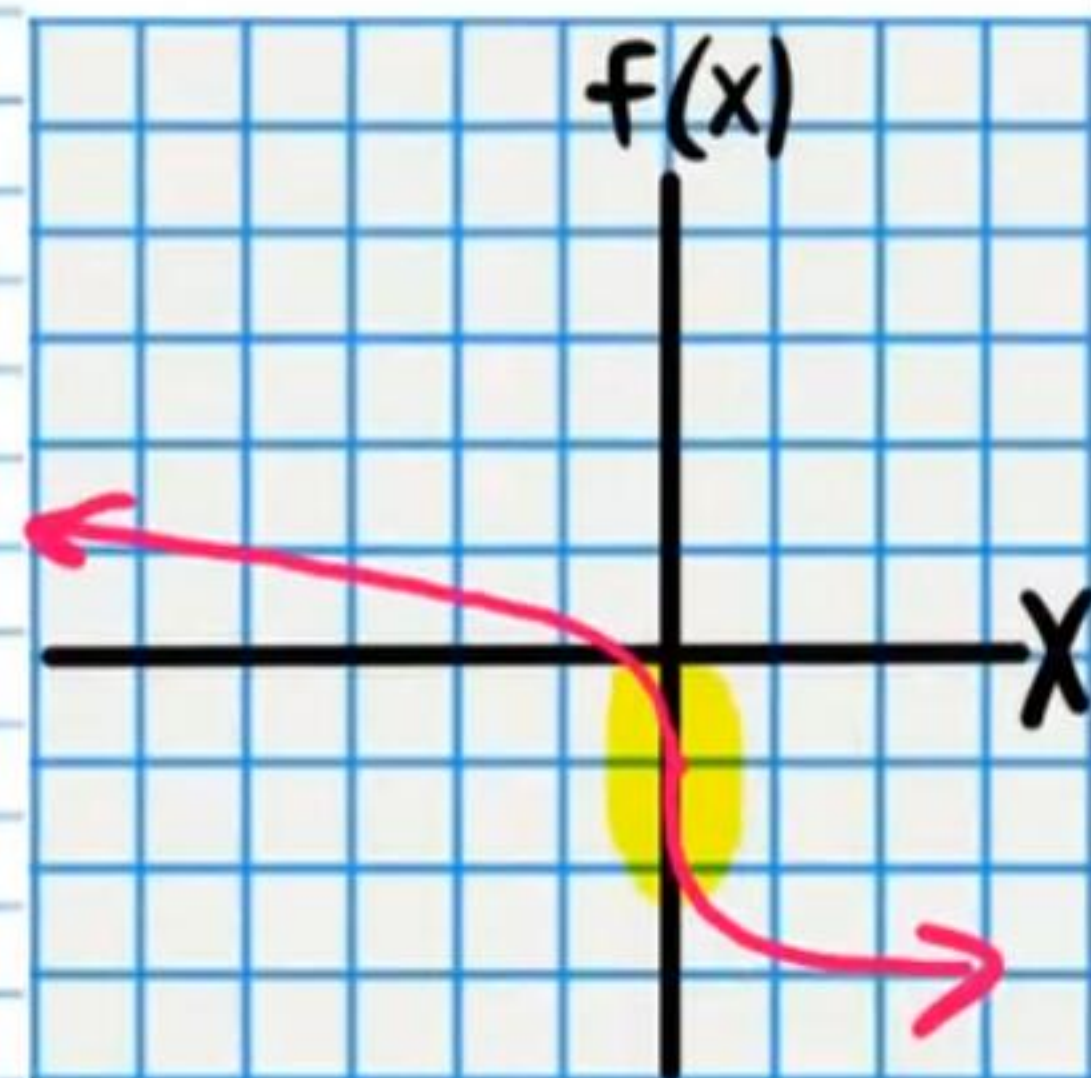
b)

Demonstrate algebraically that each function below is not differentiable at the indicated x-value (a graph of each function is shown at the right).

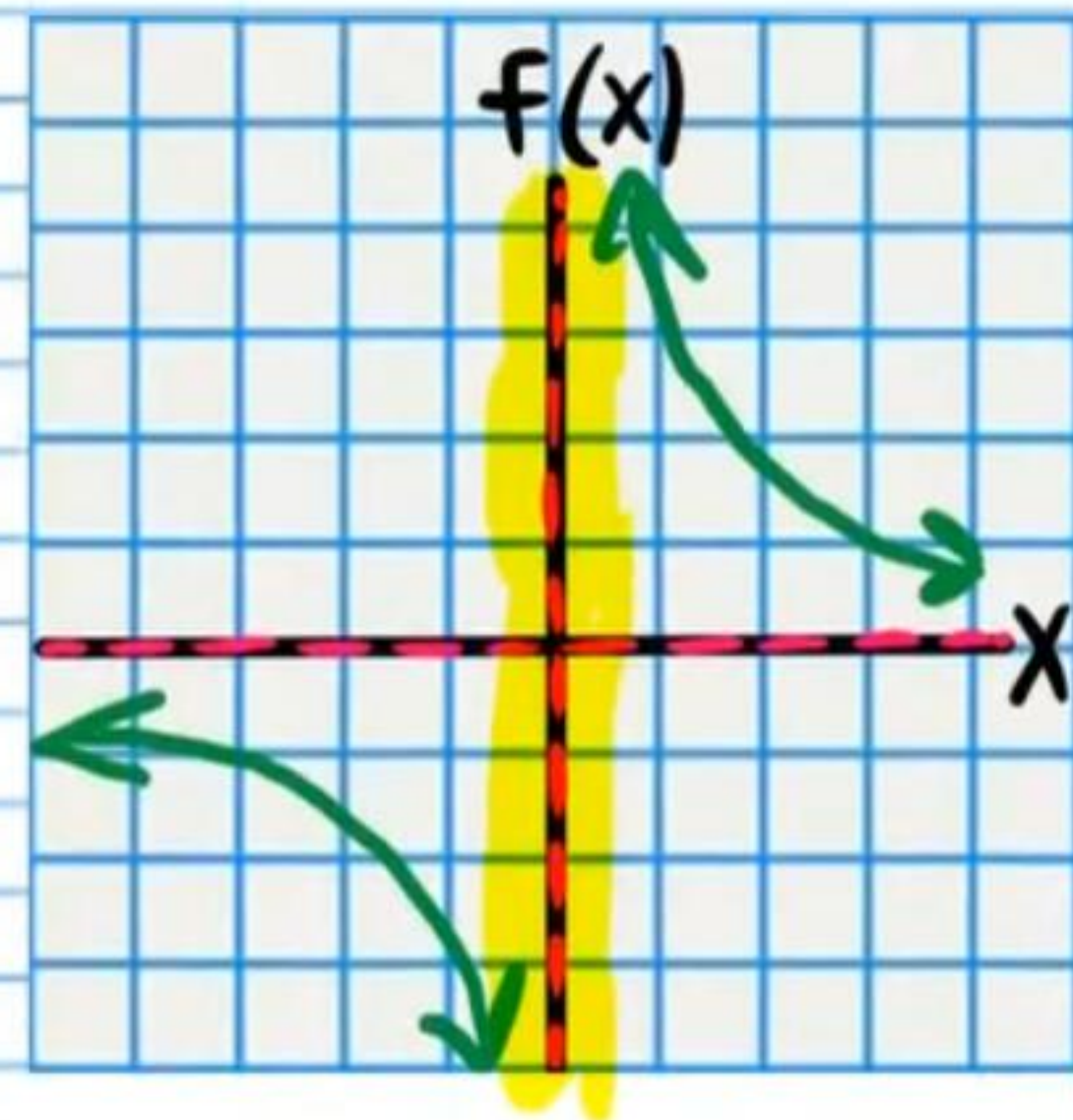
⑲ $f(x) = (x-3)^{\frac{4}{5}}$, $x=3$



②② $f(x) = -\sqrt[3]{2x} - 1$, $x=0$



②① $f(x) = \frac{3}{x}$, $x=0$



Homework Answers

① Domain

Words: All real numbers between -6 and -2, including -6.

Set Notation: $\{x | -6 \leq x < -2\}$

Interval Notation: $[-6, -2)$

Range

Words: All real numbers between 2 and 4, including 2.

Set Notation: $\{y | 2 \leq y < 4\}$

Interval Notation: $[2, 4)$

② Domain

Words: All real numbers between -3 and 5, including 5.

Set Notation: $\{x | -3 < x \leq 5\}$

Interval Notation: $(-3, 5]$

Range

Words: All real numbers between -3 and 5, inclusive.

Set Notation: $\{y | -3 \leq y \leq 5\}$

Interval Notation: $[-3, 5]$

③ Domain

Words: All real numbers greater than or equal to -5.

Set Notation: $\{x | x \geq -5\}$

Interval Notation: $[-5, \infty)$

Range

Words: All real numbers greater than or equal to -2.

Set Notation: $\{y | y \geq -2\}$

Interval Notation: $[-2, \infty)$

④ Domain

Words: All real numbers between -6 and -3, inclusive, or all real numbers greater than 1.

Set Notation: $\{x | -6 \leq x \leq -3 \text{ or } x > 1\}$

Interval Notation: $[-6, -3] \cup (1, \infty)$

Range

Words: All real numbers less than or equal to -1.

Set Notation: $\{y | y \leq -1\}$

Interval Notation: $(-\infty, -1]$

⑤ Domain

Words: All real numbers except -4 .

Set Notation: $\{x \mid x \neq -4\}$

Interval Notation: $(-\infty, -4) \cup (-4, \infty)$

Range

Words: All real numbers between -4 and -1 , inclusive.

Set Notation: $\{y \mid -4 \leq y \leq -1\}$

Interval Notation: $[-4, -1]$

⑥ Domain

Words: All real numbers except -3 .

Set Notation: $\{x \mid x \neq -3\}$

Interval Notation: $(-\infty, -3) \cup (-3, \infty)$

Range

Words: All real numbers except 1 .

Set Notation: $\{y \mid y \neq 1\}$

Interval Notation: $(-\infty, 1) \cup (1, \infty)$

⑦ Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

Range

Words: All real numbers greater than or equal to -2 .

Set Notation: $\{y \mid y \geq -2\}$

Interval Notation: $[-2, \infty)$

⑧ Domain

Words: All real numbers except -2 and 3 .

Set Notation: $\{x \mid x \neq -2 \text{ and } x \neq 3\}$

Interval Notation: $(-\infty, -2) \cup (-2, 3) \cup (3, \infty)$

Range

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

⑨ Domain

Words: All real numbers except $\frac{1}{3}$.

Set Notation: $\{x \mid x \neq \frac{1}{3}\}$

Interval Notation: $(-\infty, \frac{1}{3}) \cup (\frac{1}{3}, \infty)$

⑩ Domain

Words: All real numbers greater than or equal to 6.

Set Notation: $\{x \mid x \geq 6\}$

Interval Notation: $[6, \infty)$

⑪ Domain

Words: All real numbers less than 15.

Set Notation: $\{x \mid x < 15\}$

Interval Notation: $(-\infty, 15)$

⑫ Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

⑬ Domain

Words: All real numbers.

Set Notation: All real numbers.

Interval Notation: $(-\infty, \infty)$

⑭ a) None
b) $x = -3, x = -2, x = 2$

⑮ a) $x = -3$
b) $x = -3$

⑯ a) $x = -2, x = 2$
b) $x = -2, x = 2$

⑰ a) $x = 1$
b) $x = -1, x = 1, x = 3$

⑱ a) $x = -1$
b) $x = -3, x = -1, x = 1$

⑲ $f'(x) = \frac{4}{5\sqrt[5]{x-3}}$ $f'(3)$ is undefined.

⑳ $f'(x) = -\frac{1}{3\sqrt[3]{(2x)^2}}$ or $-\frac{1}{3\sqrt[3]{4x^2}}$ $f'(0)$ is undefined.

② $f'(x) = -\frac{3}{x^2}$ $f'(0)$ is undefined.